ISSN 2412-8317
Refereed no. Twenty five

Alamana

Semi- annual refereed scientific journal issued by Studies and Researches Center at the Sultan Qaboos Academy for Police Sciences

المؤتمر العلمي الدولي الثالث
لأكاديمية السلطان قابوس لعلوم الشرطة
الجرائم العابرة للحدود
التحديات والاستراتيجيات وتقنيات المواجهة

Part three

Predictive Artificial Intelligence in Border Security: A Criminological Frame-work for Countering Evolving Transnational Threats

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The modern framework of global border security is experiencing a significant structural transformation. Contemporary nation-states are transitioning from fixed, geographically defined borders to fluid, data-informed security boundaries. Geographical boundaries have evolved beyond simple lines of separation or localised checkpoints; they are becoming decentralised, "expanding digital filters" driven by artificial intelligence (AI), predictive analytics, and automated sensor networks. In a time marked by extreme globalisation and swift technological advancement, the effectiveness of border security depends significantly on a nation's capacity to absorb, analyse, and implement extensive flows of real-time intelligence well in advance of any threat arriving at a physical entry point.

For the Sultanate of Oman, this technology advancement is not only an operational enhancement—it is a strategic imperative closely tied with the national objectives of Oman Vision 2040. As the Sultanate hastens its efforts in economic diversification, enhances its maritime trade facilities, and establishes itself as a leading global logistics center linking Asia, Europe, and Africa, the amount of cross-border human and commercial movement will increase dramatically. Maintaining national sovereignty, public safety, and economic stability necessitates an updated security system that can promote lawful trade and mobility while effectively detecting illegal activity.

Revamping the operational structure of the Royal Oman Police (ROP) with cutting-edge predictive technology aligns with the fundamental tenets of Vision 2040 concerning institutional governance, national resilience, and technological advancement. By integrating predictive AI into its border infrastructure, Oman enhances its status as a

secure, stable, and technologically advanced state capable of minimising modern security threats with high efficiency and scientific precision.

1.1 The primary obstacle facing modern border control agencies is the disparity between the operational flexibility of Transnational Organised Crime (TOC) syndicates and the conventional methods of national security. Criminal organisations involved in people trafficking, drug smuggling, arms trafficking, and maritime contraband have swiftly adjusted to technological progress. These illicit organisations function with considerable adaptability, non-hierarchical leadership frameworks, and advanced evasion tactics tailored to capitalise on deficiencies in standard border monitoring.

A significant obstacle to efficient law enforcement is the extensive "dark figure" of crime, the considerable amount of unlawful border crossings, illegal transactions, and sea smuggling activities that remain completely unnoticed or unrecorded by conventional security measures. Traditional border security frameworks are inherently reactive; they depend significantly on human-operated inspections, manual document assessments, and physical patrols that initiate responses post-violation or breach.

In densely populated transit areas—like bustling land borders, international airports, and critical marine bottlenecks—human border agents encounter significant data overload. Human cognitive ability is intrinsically constrained while managing thousands of daily manifests, passenger name records (PNR), and ongoing surveillance streams. This cognitive strain results in operational impediments: officers are compelled to either do laborious manual verifications that hinder legal commerce or accelerate processing, compromising comprehensive risk assessment. As a result, reactive approaches create significant security weaknesses that organised crime syndicates routinely take advantage of.

1.2 This research examines how predictive AI can convert border management from a reactive, containment-oriented approach to a proactive, deterrence-based framework to rectify this significant operational disparity. Instead than reacting to breaches post-occurrence, predictive solutions utilise sophisticated algorithms to anticipate likely threat vectors, detect irregularities in extensive datasets, and implement security

measures proactively.

The principal aims of this study are outlined as follows:

Assess Predictive Frameworks:

1. To attain these goals, this research examines the subsequent fundamental enquiries:

Principal Inquiry Query: In what manner may predictive Artificial Intelligence function as a "digitally adept protector" to bolster national border resilience and advance proactive danger identification in the Sultanate of Oman, in accordance with the objectives of Oman Vision 2040?

Secondary Inquiry Questions: What are the primary operational constraints of existing reactive border security frameworks in addressing contemporary transnational organised criminal syndicates?

In what ways may sophisticated confronting technologies—namely Machine Learning (ML) for behavioural pattern recognition and Geospatial Intelligence (GEOINT)—be incorporated into the operational processes of the Royal Oman Police (ROP)?

What technical, ethical, and criminological governance frameworks are necessary inside the Omani legal system to avert algorithmic bias, safeguard information sovereignty, and adhere to UNODC standards?

1.3 The integration of developing technology to combat intricate transnational security concerns has become a major focus of recent research writing on border management. Three key theme domains can be used to broadly classify current research:

Automated Surveillance and Sensor Integration: The use of autonomous drone swarms, multi-agent reinforcement learning, and machine learning models for real-time perimeter monitoring has been thoroughly studied by researchers like Open Research Europe (2026) and Semantic Scholar Research (2026). The technical effectiveness, detection precision, and algorithmic capabilities of hardware-software integration over land and maritime boundaries are the main topics of these investigations.

Data-Driven Risk Profiling: Studies conducted by SITA (2023) and Travizory (2024) demonstrate the practical use of advanced passenger processing systems, such as Passenger Name Record (PNR) analysis and Advanced Passenger Information (API). Predictive analytics can expedite passenger verification and detect high-risk anomalies before passengers physically arrive at entrance checks, as this body of work shows.

Ethical, Legal, and Governance Frameworks: The systemic concerns connected to the use of AI in law enforcement are covered in an expanding body of literature. The risks of automated bias, black-box decision-making, and the deterioration of human oversight in high-stakes security operations are highlighted by authors concentrating on algorithmic accountability (Decision Lab, 2024; EIMT, 2026) and institutional policy frameworks (IRIS, 2026; UNODC, 2023).

The Gap in Research

There are still two major gaps in literature, despite the fact that it provides solid insights into either broad ethical criticisms or pure technical implementations:

1.Theoretical Synthesis: There is a dearth of criminological literature that specifically updates conventional ideas (like the "capable guardian") into digital and algorithmic contexts specific to border surveillance by explicitly integrating predictive AI frameworks within classical routine activity theory (RAT).

2.Contextual Application: Predictive AI frameworks are rarely translated into practical, operational roadmaps that are adapted to the Sultanate of Oman's and the Gulf region's strategic, geographical, and legal architecture.

This research directly overcomes these gaps by presenting an integrated criminological paradigm that combines predictive AI deployment with practical feasibility and solid human-in-the-loop safeguards.

1.4 This study focuses on the strategic, criminological, and legal parameters of employing predictive AI for border security within the Sultanate of Oman. The study operationally tackles the unique geographic and environmental challenges faced by Oman, encompassing:

High-Density Land Borders: Essential entrance points like Al Wajajah necessitate swift yet comprehensive inspections for both commercial and passenger traffic.

Rugged Landscape: The extensive, thinly inhabited desert regions (Rub' al Khali) and the elevated topography of the Dhofar Governorate, which pose significant problems for physical monitoring.

Strategic Maritime Routes: Oman possesses a vast coastline and essential maritime bottlenecks, particularly the Strait of Hormuz, necessitating ongoing, unobtrusive maritime domain surveillance.

Methodological Limitations:

This study constitutes an autonomous scholarly investigation designed for a peer-reviewed scientific conference, drawing solely on publicly available literature, published policy frameworks (like Oman Vision 2040), international legal structures (UNODC), and declassified comparative assessments of global border systems (Frontex). The research does not employ classified operational information, confidential state secrets, or restricted intelligence documents from the Royal Oman Police. All framework suggestions are crafted as overarching conceptual and policy-focused models aimed at guiding future research and strategic development.

1.5 This research is methodically organised into seven interrelated chapters:

Chapter 1: Introduction — Sets the framework, identifies the research issue, outlines objectives, formulates research enquiries, defines the scope, and aligns with Oman Vision 2040.

Chapter 2: Methodology & Theoretical Framework — Details the qualitative policy-analysis methodology, the comparative case study method, and the utilisation of Routine Activity Theory (Cohen & Felson, 1979) to characterise AI as a "digitally proficient guardian.

Chapter 3: Theme 1 — Strategic and Ethical Dilemmas in Contemporary Border Management — Analyses the adaptability of Transnational Organised Crime, algorithmic prejudice, data sovereignty, and the necessity of human oversight.

Chapter 4: Theme 2 — Confrontation Technologies: The Digital Guardians — Explores the technical intricacies of Machine Learning (ML) risk assessment, GEOINT, UAV systems, and instantaneous predictive modelling.

Chapter 5: Theme 3 — Implementing Counterstrategies: Insights from Frontex for the ROP — Undertakes a comparative analysis of Europe's Frontex effort, deriving operational insights and tailoring them to Oman's legal and geographical framework.

Chapter 6: Synthesis: The Suggested Omani Criminological Governance Structure Outlines the structural blueprint of the proposed Omani "Smart Border Framework" and its conformity with UNODC (2023) criteria.

Chapter 7: Conclusion & Policy Recommendations – Encapsulates principal discoveries, offers implementable guidance for Omani security authorities, and delineates prospective research avenues.

1. ***This research utilises a non-empirical, qualitative, theory-based policy analysis combined with an international comparative case study approach. Given the strategic nature of state border security infrastructure and the sensitive classification surrounding operational law enforcement protocols, an empirical quantitative design relying on live police databases or classified agency feeds is neither feasible nor legally permissible for an open academic inquiry.***

A qualitative policy-analysis approach facilitates a thorough structural evaluation of the conceptual,

legal, and operational aspects of predictive Artificial Intelligence (AI) in border control. The research framework is organised into two evaluative stages:

Theoretical Inference: Utilising recognised principles of environmental criminology—particularly Routine Activity Theory (Cohen & Felson, 1979)—to understand how automated prediction software, Machine Learning (ML), and Geospatial Intelligence (GEOINT) alter spatial and situational deterrence.

Comparative Policy Analysis: Conducting a systematic comparative evaluation between an established international baseline—the European Border and Coast Guard Agency

(Frontex) "Smart Borders" initiative—and the operational requirements of the Royal Oman Police (ROP) under the governance goals of Oman Vision 2040.

This framework produces adaptable strategic insights by integrating theoretical reasoning with comparative policy analysis, all while upholding stringent academic standards.

1.1 This research employs a multi-layered qualitative data set to ensure strong themed validity and analytical rigour. The review links five main types of reliable, open-access information to create a broad base for policy analysis:

Global Legal and Regulatory Structures: Publications from international governance bodies, mainly the United Nations Office on Drugs and Crime (UNODC), including treaties on transnational organized crime (like the Palermo Convention), guidelines on cross-border data protection, and procedural manuals for integrated border management.

National Strategic Blueprints: Authoritative policy and legislative documents of the Sultanate of Oman, mainly Oman Vision 2040, national legislative mandates about cyber governance and data sovereignty, and strategic directives for the modernization of the Royal Oman Police (ROP) and national security infrastructure.

Assessed international assessments: Formal evaluation reports, operational analyses, legal assessments, and fundamental rights assessments of European automated border technology implementations, particularly published assessments of the European Travel Information and Authorization System (ETIAS), the Entry/Exit System (EES), and Frontex situational awareness frameworks.

Peer-Reviewed Scholarly Literature: Scholarly academic articles and theoretical texts on environmental criminology, Routine Activity Theory (RAT), situational crime prevention, predictive policing algorithms, geospatial surveillance technologies, and algorithmic ethics in the criminal justice system.

Technical White Papers and Industry Reports: Specialized technical documentation and operational briefs by established worldwide border security technology providers (e.g., SITA, Travizory) and applied AI research institutes (e.g., Decision Lab, EIMT). while primarily relied on peer-reviewed literature and official policy but also intentionally included these technical sources to capture real-world operational deployments, system architectures, and practical industry norms for API/PNR analytics and AI risk-scoring mechanisms that are not yet fully documented in traditional academic journals.

1.2 Data gathering adhered to a methodical document-filtering procedure to extract highly pertinent, credible literature while discarding conjectural or non-scholarly opinion. The selection process was carried out in three consecutive stages:

Phase One: Keyword Exploration and Database Interrogation: Thorough investigations were performed throughout global scientific databases (e.g., Scopus, Web of Science, Google Scholar) and organisational archives (UNODC, EU Publications Office, Omani Ministry of Legal Affairs). Search criteria amalgamated through boolean searches including: ("Predictive AI" OR "Machine Learning") AND ("Border Security" OR "Transnational

Organised Crime") AND ("Routine Activity Theory" OR "Situational Crime Prevention").

Phase 2: Inclusion and Exclusion Screening: Acquired documents were evaluated based on established methodological standards. Sources must meet the following criteria: (a) published by recognised governmental, international, or scholarly institutions; (b) pertinent to border security technology, criminological theory, or policies of the Omani state; and (c) released within a period that reflects contemporary AI developments while adhering to fundamental criminological principles. Speculative news pieces and unsubstantiated technical blogs were methodically omitted.

Phase 3: Thematic Synthesis & Data Extraction: Chosen materials were classified into structured analytical frameworks centred on three principal thematic categories: Operational Efficiency, Technical and Legal Deficiencies, and Algorithmic Ethics and Governance. This structural framework guaranteed that data obtained from European comparative instances could be methodically compared with Omani legal and geographical contexts.

1.3 The essential analytical framework of this research is based on Routine Activity Theory (RAT), initially developed by Marcus Felson and Lawrence E. Cohen in 1979.

Conventional Routine Activity Theory asserts that a criminal incident transpires when three fundamental components coalesce concurrently in both space and time:

Criminal Event=Motivated Offender+Suitable Target+Absence of a Capable Guardian

In traditional border enforcement, this triangle functions in a tangible manner: a determined smuggling organization locates an unguarded border area (appropriate target/vulnerability) where border officers are either missing or preoccupied (lack of an effective guardian), facilitating unauthorised access.

Disassembling the Triad in the Age of Predictive AI

When implemented in contemporary, technology-oriented border management, the fundamental principles of RAT experience a structural alteration:

The Driven Criminal: Transnational organised crime (TOC) networks, human traffickers, and illicit goods smugglers. In the modern landscape, these entities consistently modify their timing, pathways, and evasion techniques in response to identified weaknesses in border monitoring.

The Appropriate Focus: Oman's border facilities, comprising high-capacity land ports (such as Al Wajajah), extensive desert frontiers next to the Empty Quarter, and important maritime access points along the Sea of Oman and the Strait of Hormuz. The allure of these targets to criminal organisations escalates when processing checkpoints encounter data overload and personnel exhaustion.

Reconceptualising the "Capable Guardian": Historically, a guardian was only characterised as a tangible presence—such as a law enforcement officer, customs official, or physical obstruction. This research reconceptualises Machine Learning (ML) prediction systems and Geospatial Intelligence (GEOINT) frameworks as "Digitally Proficient Custodians.

Mitigating Spatial Convergence

In contrast to human guardians who respond when an offender and target intersect, a Digitally Competent Guardian functions proactively. Through the amalgamation of ongoing risk-assessment algorithms and instantaneous GEOINT surveillance (such autonomous UAV sensor operations over isolated landscapes), predictive AI discerns nuanced irregularities and spatial trends well in advance of physical convergence.

By predicting high-likelihood threat zones, the system guides human border agents to preemptively address threats before a breach occurs. Essentially, predictive AI transforms the geographical opportunity framework of criminal activity, changing the former "lack of a guardian" into an ever-present digital filter.

1.4 Connecting fundamental computer science

concepts (such as algorithms and neural networks) with practical law enforcement tactics necessitates a robust criminological structure. Analyses of AI that are strictly technical frequently emphasise computing precision, processing velocity, or server capability—neglecting the operational, legal, and human elements that influence the real-world efficacy of a security technology.

Utilising Routine Activity Theory (RAT) in formulating state technology strategy offers numerous strategic benefits:

Operational Translation: It converts intricate algorithmic results (such probabilistic threat assessments and heat maps) into practical law enforcement ideas that police leaders and field officers in the Royal Oman Police may promptly implement.

Strategic Congruence with Oman Vision 2040: It offers an organised framework to assess how technical expenditures directly meet governmental aims of institutional efficacy, threat mitigation, and contemporary governance.

Mitigating Technological Determinism: By integrating AI tools into situational crime prevention theory, the study guarantees that technology is perceived not as a total substitute for human discernment, but as an augmentative force that amplifies institutional oversight.

1.5 Engaging in research with state security and predictive technologies necessitates rigorous compliance with ethical research standards and national data protection regulations.

Scholarly Honesty & Non-Violation: This research is exclusively founded on publicly available, unclassified sources. It refrains from attempting to access, rebuild, or assess proprietary algorithms, internal police databases, or protected state secrets of the Royal Oman Police.

Data Sovereignty: The theoretical models presented in this research emphasise complete governmental control over data. In accordance with Omani legislative regulations, any suggested digital framework asserts that critical national security information must be solely governed by the Omani government, safeguarded against illicit external intrusion or global data acquisition.

Algorithmic Ethics and Human Rights: The methodological framework expressly integrates ethical safeguards to mitigate the dangers of "algorithmic prejudice. The evaluative structure examines methods to guarantee that predictive risk-assessment instruments adhere to global human rights norms (UNODC) and Omani constitutional law, ensuring that automated danger evaluations are consistently monitored by human oversight ("Human-in-the-loop").

2. ***The primary operational challenge in modern border security is the fundamental transformation in the functioning of transnational organised crime (TOC) networks. Historically, illicit organisations operated via strict, hierarchical command frameworks***

with fixed transportation pathways. Contemporary criminal organisations have transformed into decentralised, agile, and technologically adept networks that consistently adjust to circumvent conventional governmental protections (United Nations Office on Drugs and Crime [UNODC], 2023, 2024).Instead of immediately facing fixed physical obstacles or established access points, criminal organisations take advantage of structural "friction points" within governmental security frameworks. These organisations employ encrypted communication systems, darknet platforms, commercial counter-surveillance equipment, and sophisticated logistical strategies to transport illegal goods, drugs, arms, and victims of human trafficking across borders (UNODC, 2024).

These networks flourish by using the "dark figure" of crime, which encompasses the large amount of illicit activities that go untracked by traditional law enforcement. When state law enforcement depends solely on fixed physical structures and routine human patrols, criminal organisations can readily identify operational vulnerabilities. Mitigating such nimble threats necessitates that border security personnel

evolve from static, reactive stances to proactive, intelligence-driven operations adept at anticipating criminal movement.

2.1 The implementation of automated threat detection and predictive AI presents a significant structural dilemma for state governance: reconciling operational efficiency with national information sovereignty and judicial autonomy.

The Operational Dilemma of Data Evaluation

High-density land checkpoints and significant maritime ports handle substantial quantities of commercial freight, passenger lists and travel documentation on a daily basis. In order to avert significant economic impediments, law enforcement bodies are under considerable pressure to enhance scanning efficacy via automated algorithms (Outram, 2025).

Nevertheless, excessive dependence on external, third-party, or cross-border automated screening mechanisms poses considerable threats to information sovereignty. National security information—comprising biometric records, travel logs, and live movement data—represents essential governmental resources. Allowing opaque algorithms created by external software providers to handle national security data without domestic supervision may jeopardise governmental authority, risking the exposure of confidential internal intelligence to global data collection or foreign regulatory intrusion (Filho, 2025; Gugliotta, 2023).

Alleviation of Automation Bias: A significant operational risk linked to automated threat-scoring systems is automation bias, which refers to the psychological inclination of human operators to excessively rely on automated results and unquestioningly accept machine suggestions, despite the presence of conflicting field evidence (Decision Lab, 2024; EIMT, 2026).

In border security contexts, automation bias poses a dual risk:

Erroneous Alerts: Law enforcement may disproportionately focus on low-risk individuals identified by an algorithm only due to "the system issuing a notification," thereby squandering essential operational resources and violating civil liberties (EIMT, 2026).

False Negatives: Law enforcement personnel can disregard authentic, significant security risks due to an automated screening system not issuing an alert, so fostering a misleading sense of safety (EIMT, 2026).

To maintain the autonomy of the Omani legal system, prediction instruments should never function as independent adjudicators. The Royal Oman Police (ROP) should enforce a rigorous "Human-in-the-Loop" (HITL) operational framework. In accordance with this procedure, predictive AI serves solely as an analytical advisory instrument, guaranteeing that ultimate legal evaluations and physical interventions are retained under the explicit jurisdiction and expertise of qualified Omani border officials (Babuta & Oswald, 2019; European Commission, 2024).

2.2 The incorporation of Machine Learning (ML) inside law enforcement raises substantial criminological and ethical issues pertaining to algorithmic bias. In opposition to common beliefs that automated algorithms possess intrinsic objectivity, machine learning models are developed using past data. If past law enforcement data exhibits systemic deficiencies, regional over-policing, or previous demographic disparities, the machine learning algorithm will assimilate, entrench, and exacerbate these shortcomings (AlgorithmWatch, 2025; Barman & Barman, 2026).

Processes of Algorithmic Bias

In predictive border enforcement, prejudice often emerges via computerised risk assessment. If an algorithm is developed using historical arrest data that disproportionately represents certain travel patterns or nationalities—rather than genuine crime statistics—the system will produce exaggerated threat assessments for individuals with those demographic characteristics. This establishes a self-perpetuating feedback cycle:

The program identifies a particular demographic vector as "high risk.

Border officials predominantly focus their inspection activities on that avenue

Authorities identify a greater number of minor violations within that cohort owing to heightened examination.

The updated arrest statistics are reintegrated into the system, so reinforcing the algorithm's prejudiced presumption (AlgorithmWatch, 2025; Babuta & Oswald, 2019).

Alleviating Unequal Effects According to Omani Legislation

To guarantee that predictive instruments enhance—rather than undermine—the integrity of Omani law enforcement, risk-assessment frameworks must be designed to avert disparate effects. Predictive systems implemented by the ROP shall avoid employing sensitive protected characteristics (such as ethnicity, nationality, religious affiliation, or arbitrary demographic indicators) as principal factors in threat-assessment algorithms

(AlgorithmWatch, 2025). Algorithms should concentrate only on identifying behavioural anomalies and objective spatial markers—such as atypical travel patterns, abrupt route changes, inconsistencies in cargo weight, or unauthorised boundary breaches identified by sensors (Outram, 2025).

Harmonising AI enhancement with Omani constitutional tenets and Islamic legal concepts necessitates:

Algorithmic Clarity: Guaranteeing that AI threat models are entirely verifiable by Omani legal and security entities, hence removing incomprehensible "black-box" determinations.

Ongoing Ethical Auditing: Conducting regular data assessments to ensure that predictive algorithms do not produce consistent discriminatory results against particular authorised traveller groups.

Adherence to Oman Vision 2040: Maintaining elevated institutional governance benchmarks through the integration of technical advancements and rigorous commitment to human rights, legal frameworks, and national sovereignty

3. ***The fundamental mechanism of proactive border management depends on replacing responsive human examination with rapid automated data analysis (Outram, 2025; SITA, 2023). In situations with significant border traffic, such as international airports, maritime ports, and busy land crossings, the influx of travel and commerce data surpasses human analytical capabilities (Kostov, 2021).***

To surmount these cognitive constraints, contemporary confronting technologies utilise supervised classification algorithms and anomaly detection models on essential datasets (Kostov, 2021; Travizory, 2024):

Passenger Name Records (PNR): Encompassing travel schedules, payment options, luggage handling, contact information, and seat allocation trends (Kostov, 2021; SAS Institute, 2021).

Advanced Passenger Information (API): Confirming biographical details, passport authenticity, and national watch-list conditions before departure (SITA, 2023; Travizory, 2024).

Customs Information & Cargo Documentation: Examining shipping sources, commodity masses, freight declaration records, and supply-chain routing frameworks (Outram, 2025; SITA, 2023).

Shifting to Probabilistic Threat Assessment

Historically, border screening depended on deterministic rule-based algorithms (Kostov, 2021). These conventional methods functioned based on inflexible, predetermined logic (e.g., If a traveller corresponds with watchlist entry X, then initiate inspection Y).

Although proficient in identifying recognised perpetrators, deterministic algorithms are inadequate in detecting unidentified illicit actors that deliberately arrange their journey to evade watchlists (Barrett,

2017; Kostov, 2021).

Machine Learning (ML) signifies a transition towards predictive probabilistic sampling (Kostov, 2021; Open Research Europe, 2026). Employing supervised classification algorithms (including Random Forests, Gradient Boosting, and Neural Networks) that are trained on past violation data, machine learning engines evaluate intricate behavioural patterns concurrently (Kostov, 2021; SAS Institute, 2021).

Instead of identifying overt rule breaches, probabilistic systems assess a real-time threat score for every passenger or cargo manifest utilising dynamic anomaly indicators (e.g., last-minute cash ticket acquisitions coupled with atypical transit paths and unverified contact information) (SAS Institute, 2021; Travizory, 2024). This enables border control authorities to recognise "unknown-unknown" threats—highlighting high-probability

illegal pathways prior to the arrival of the person or goods at the customs counter (Barrett, 2017; Kostov, 2021).

3.1 Data analytics evaluates organised information at designated entry points, but safeguarding undefined borders necessitates sophisticated spatial confrontation technologies (Lugga, 2025). Oman's geography—characterized by the extensive desert regions of the Rub' al Khali (Empty Quarter), the formidable mountain ranges of Dhofar, and crucial coastal areas along the Sea of Oman and the Strait of Hormuz—features extensive uninhabited landscapes where physical border checkpoints cannot be perpetually staffed.

Independent UAV Swarms and Advanced Reinforcement Learning

Ensuring the protection of these extensive, isolated boundaries necessitates the amalgamation of Geospatial Intelligence (GEOINT) with Unmanned Aerial Vehicles (UAVs) (Lugga, 2025; Semantic Scholar Research, 2026). Conventional drone operations depend on fixed flight trajectories or direct manual control, which are hindered by operator tiredness and restricted battery life (Punjab State Open University, 2026; Semantic Scholar Research, 2026).

Contemporary border security substitutes traditional flight methods with Deep Reinforcement Learning (DRL) algorithms (MDPI Sensors, 2023; Semantic Scholar Research, 2026). Autonomous UAV swarms, functioning within multi-agent reinforcement learning (MARL) paradigms, adjust their flight paths in response to immediate environmental data (Semantic Scholar Research, 2026):

Proactive Route Enhancement: Deep Reinforcement Learning models incessantly assess geographical characteristics, meteorological conditions, and past smuggling locations to refine search zones autonomously (MDPI Sensors, 2023; Semantic Scholar Research, 2026).

Adaptive Target Surveillance: Upon detecting illicit activity in isolated landscapes, the UAV swarm independently modifies its flight trajectories to ensure uninterrupted visual monitoring while notifying ground units (Punjab State Open University, 2026; Semantic Scholar Research, 2026).

Multi-Sensor Integration as a "Digitally Proficient Custodian To eradicate coverage deficiencies, GEOINT systems integrate information from many sensory modalities (IRIS, 2026; Lugga, 2025):

High-Resolution Satellite Imagery: Facilitating extensive terrain mapping and observing structural alterations in isolated border areas (IRIS, 2026; Lugga, 2025).

Thermal and Infrared Imaging: Identifying heat signatures of illicit vehicles or individuals navigating desert or hilly paths during nighttime (Lugga, 2025).

Ground-Installed Acoustic and Seismic Networks: Detecting footstep patterns or low-flying drone incursions adjacent to vital perimeter boundaries (Lugga, 2025; Punjab State Open University, 2026).

Reassessing this technical network via Routine Activity Theory (Cohen & Felson, 1979) uncovers its criminological role. In isolated border areas,

criminal organisations exploit the assumed "lack of an effective overseer" to carry out illicit crossings without detection (Cohen & Felson, 1979; Monash University, 2025). Multi-sensor GEOINT integration functions as an ongoing "Digitally Proficient Sentinel," eliminating terrain-induced obscurity and hindering illicit activities in sparsely populated areas (Lugga, 2025; Monash University, 2025).

3.2 The integration of real-time machine learning risk assessment with GEOINT sensor data facilitates predictive modelling (Lugga, 2025; Outram, 2025). Instead of presenting unchanging historical information on control room screens, sophisticated command settings employ dynamic digital depictions of border ecosystems (IRIS, 2026; Lugga, 2025). prediction simulation engines analyse ongoing flows of operational data—including meteorological changes, maritime tidal conditions, satellite thermal irregularities, and past smuggling timelines—to execute real-time prediction simulations (Lugga, 2025; Punjab State Open University, 2026). These simulations produce predictive threat heat maps that anticipate the timing, location, and manner of a potential border breach during the forthcoming 12 to 48 hours (Lugga, 2025; Outram, 2025). This capacity shifts the operational approach of the Royal Oman Police (ROP) from reactive deployment to proactive placement (Lugga, 2025; Outram, 2025). By predicting criminal patterns prior to actual incursions, tactical leaders can strategically locate patrols, utilise interception drones, and modify checkpoint inspection procedures in advance (Lugga, 2025; Punjab State Open University, 2026). This strategy eliminates the time frame that criminal organisations depend on, attaining situational deterrence via digitally orchestrated law enforcement (Cohen & Felson, 1979; Monash University, 2025).

4. The European Border and Coast Guard Agency (Frontex) and the overarching border management system of the European Union exemplify one of the most comprehensive applications of automated, data-centric border infrastructure globally. Assessing the operational path of the EU's "Smart Borders" initiative offers vital insights for the Royal Oman Police (ROP) as Oman enhances its digital border framework.

4.2.1 The primary accomplishment of the EU Smart Borders initiative is its systemic interoperability. By means of legislative and technical amalgamation, the EU interconnected formerly segregated databases—particularly the European Travel Information and Authorisation System (ETIAS), the Visa Information System (VIS), and the Entry/Exit System (EES)—into cohesive central query frameworks like the Common Identity Repository (CIR).

This compatible framework produces considerable operational advantages:

Expedited Pre-Screening: ETIAS autonomously evaluates millions of visa-exempt travel requests against global watchlists and risk-assessment criteria before passenger departure, distinguishing low-risk travellers while promptly identifying high-risk irregularities.

Cross-Database Querying: Interoperability allows border officials to concurrently cross-check biographical, biometric, and travel history information across various regional systems during manual inspections, thereby thwarting identity fraud and alias exploitation.

Automated Overstay Monitoring: The EES digitally logs entry and exit times, eliminating the need for physical passport stamps and promptly recognising persons who beyond their authorised stay periods.

4.2.2 Notwithstanding elevated operational efficacy, the implementation of Frontex and EU intelligent border instruments faced considerable legal, institutional, and technical obstacles: Systemic Obscurity ("Black-Box" Processing): Automated threat-assessment algorithms scrutinise travellers employing concealed, pre-established profiling criteria. When an applicant is denied by automated profiling, the unclear workings of the risk-assessment system complicate the articulation

of specific legal justifications for the rejection, resulting in legal disputes about judicial review and due process.

The amalgamation of extensive biometric databases has perpetuated ongoing legal disputes concerning the General Data Protection Regulation (GDPR) and the EU Charter of Fundamental Rights. Concerns focused on function creep—where security data collected for border control was subsequently utilised for broader law enforcement without clear legal authorisation.

Unwritten Guidelines and Computational Mistakes: In reality, trivial programming inaccuracies or defective risk metrics in automated profiling systems consistently misidentified legitimate travellers as significant risks. Frontline border guards frequently lacked the requisite technical skills to analyse machine outputs, leading them to habitually accept automated inaccuracies and revealing inherent deficiencies in accountability.

4.1 To harness the advantages of the Frontex model while circumventing its technological and legal shortcomings, the ROP must formulate an operational framework particularly adapted to the Sultanate's administrative and legal context.

Formulating Command Procedures

The ROP ought to create a centralised Border Intelligence Fusion Center (BIFC) that consolidates real-time Geospatial Intelligence (GEOINT), Passenger Name Records (PNR), customs manifests, and maritime automated identification system (AIS) data.

In this primary facility, automated AI analytics analyse real-time data to produce visual threat heat maps and danger notifications. Nonetheless, operational procedures must implement a rigorous "Human-in-the-Loop" (HITL) operational framework:

Ingestion and Evaluation: Artificial intelligence systems perpetually monitor incoming travel and border information to assess risk metrics.

Human Verification: Any automatic notification above established risk limits is directed to a dedicated ROP Intelligence Analyst for manual assessment prior to any intervention.

Tactical Command Dispatch: Upon confirmation by a human officer, tactical commanders send field units, autonomous UAVs, or coast guard vessels straight to the designated site.

Requirements for Education and Skill Enhancement

Shifting from conventional border enforcement to a data-centric operational framework necessitates enhancing the skills of security staff. Utilising intricate data dashboards without succumbing to automation bias necessitates specialised professional training. The Sultan Qaboos Academy for Police Sciences ought to develop targeted courses emphasising:

Data Literacy and Dashboard Analysis: Educating officers to assess risk ratings critically, comprehend data origins, and detect any sensor or algorithm inaccuracies.

Upholding Professional Confidentiality: Emphasising that technology instruments yield probabilistic assessments instead of definitive legal proof, so assuring officers exercise autonomous legal discernment during on-site operations.

Technico-Legal Compliance: Instructing command staff on Omani legal regulations pertaining to data management, privacy, and evidence acquisition.

4.2 An effective smart border system cannot be directly copied from other countries; it requires customisation to align with Oman's unique geography and legal structure.

Strategic Geography: Terrestrial and Maritime Transportation Routes

Oman's border security issues are structurally distinct from those faced by mainland Europe. Oman's extensive coastline bordering the Arabian Sea, the Sea of Oman, and the Strait of Hormuz necessitates the integration of coastal radar systems, satellite monitoring, and maritime AIS data to

identify unauthorised vessel activities, illegal fishing operations, and maritime trafficking.

Rugged Land Boundaries: The steep edges of Dhofar and the vast arid landscape of the Rub' al Khali necessitate continuous spatial surveillance. Utilising deep reinforcement learning (DRL) autonomous UAV swarms and seismic ground sensors ensures surveillance of uninhabited areas where stationary border posts are unfeasible.

Legislative Modification: Harmonising AI with Omani Legislation and Islamic Legal Principles

To maintain institutional confidence and adhere to Omani legislation, automated border mechanisms must conform to the constitutional and legal traditions of the Sultanate.

Judicial Accountability and Human Responsibility: According to Omani legal doctrines and Islamic jurisprudence (Fiqh), the burden of legal accountability (Mas'uliyyah) is exclusively borne by human beings. Autonomous algorithms lack legal jurisdiction.

Implementing human supervision guarantees that each search, seizure, or border refusal is sanctioned by a properly authorised human official, preserving the integrity of court processes.

Data Sovereignty and National government: All essential databases, screening tools, and AI frameworks must be maintained on secure, state-managed Omani cloud systems, restricting foreign vendor access and aligning with the national government goals of Oman Vision 2040.

Safeguarding Against Unjust Intrusion: Pursuant to Omani legal safeguards pertaining to individual liberty and confidentiality, automated screening systems must exclusively utilise objective behavioural deviations (such as irregular travel patterns or cargo inconsistencies) instead of protected identity indicators, thereby guaranteeing equitable treatment for all bona fide travellers.

5. ***This research suggests a multi-layered Omani Criminological Governance Framework for border security to implement the theoretical, technical, and comparative findings examined in this paper. This structure combines sophisticated technological elements***

with stringent human supervision to establish a robust, anticipatory, and legally compliant border defence system.

The architectural framework operates through four successive operating levels (Lugga, 2025; Outram, 2025):

Level 1: Data Acquisition & Sensory Inputs: Integrating organised passenger information (PNR/API), commercial manifests, maritime AIS data streams, and instantaneous remote sensory inputs from satellite and autonomous UAV monitoring (Lugga, 2025; SITA, 2023).

Level 2: Sophisticated Analytics and Simulation Framework: Implementing machine learning techniques and deep reinforcement learning (DRL) models to produce probabilistic threat assessments and spatial risk simulations spanning 12 to 48 hours (MDPI Sensors, 2023; Open Research Europe, 2026).

Level 3: "Human-in-the-Loop" (HITL) Validation: Directing all high-risk automated notifications to expert analysts within the Royal Oman Police (ROP) who assess machine-generated results against legal criteria prior to sanctioning tactical measures (Babuta & Oswald, 2019; European Commission, 2024).

Level 4: Tactical Field Interdiction: Implementing precise, intelligence-driven field operations—such as deploying ground patrols, enhancing checkpoint inspections, or initiating coastal interception resources—thus eliminating the opportunities exploited by organised crime syndicates (Cohen & Felson, 1979; Monash University, 2025).

5.1 Implementing sophisticated predictive technologies necessitates rigorous compliance with the international norms set forth by the United Nations Office on Drugs and Crime (UNODC) pertaining to transnational crime prevention, data privacy, and the safeguarding of human rights (UNODC, 2023, 2024).

Enforcing Global Security Protocols

Addressing Transnational Organised Crime: The UNODC Global Report on Trafficking in Persons highlights that contemporary syndicates perpetually take advantage of border weaknesses (UNODC, 2023). Through the incorporation of automated API/PNR screening and GEOINT surveillance, the Omani framework meets its international commitments under the United Nations Convention against Transnational Organised Crime (Palermo Convention) by actively detecting human trafficking and contraband smuggling routes before they reach the port of entry (UNODC, 2023, 2024).

Data Sovereignty and Protections: In accordance with UNODC directives regarding digital evidence and global data management, all traveller profiles, biometric data, and sensory inputs gathered within Omani borders must be retained on cloud infrastructure governed by the sovereign state (Filho, 2025; NCSI, 2024). Cross-border data transfers should be conducted using official, encrypted intergovernmental protocols instead of commercial third-party avenues (Gugliotta, 2023).

Compliance with Due Process: Consistent with UNODC rule-of-law standards, automated threat profiling should avoid employing protected demographic characteristics (such as nationality, ethnicity, or religion) as primary criteria, ensuring that border interventions are solely based on substantiated behavioural irregularities (AlgorithmWatch, 2025; UNODC, 2024).

5.2 To maintain enduring institutional integrity, the Omani security framework must combine technology advancements with strong internal accountability systems. This research advocates for the formation of an autonomous Omani Security AI Governance & Oversight Committee (SAIGOC) that operates directly within the ROP command framework.Committee Duties and Supervision Guidelines These evaluations examine system outputs to confirm that automated risk-scoring algorithms do not cultivate geographical or demographic prejudices over time (Barman & Barman, 2026).

Model Comprehensibility & "Opaque System Abolition: System providers and in-house software engineers are required to demonstrate that each implemented algorithm adheres to rigorous explainability criteria (European Commission, 2024). Border officials should receive explicit, comprehensible reports detailing the rationale for a particular traveler's or cargo manifest's heightened threat assessment (EIMT, 2026; Gugliotta, 2023).

Institutional Conformity with Oman Vision 2040: The suggested framework directly bolsters Oman Vision 2040 by instituting transparent governance, comprehensive training at the Sultan Qaboos Academy for Police Sciences, and well-defined legal parameters (NCSI, 2024). It harmonises sophisticated technology defence with elevated benchmarks of public administration, national sovereignty, and judicial probity.

6. ***This study has assessed the fundamental transformation necessary to protect national borders from contemporary transnational organised crime (TOC). Conventional border security systems—dependent on fixed physical installations, routine human surveillance, and responsive enforcement—are fundamentally susceptible to decentralised, technologically adept criminal organisations that function across undefined areas and manipulated supply chains (UNODC, 2023, 2024).***

The principal conclusions of this research are that:

Confrontation Technologies Provide Tactical Advantage: The incorporation of Machine Learning (ML) probabilistic risk assessment within Passenger Name Records (PNR), Advanced Passenger Information (API), and cargo manifests allows border agencies to detect "unknown-unknown" threats prior to their physical arrival at entry points (Barrett, 2017; Kostov, 2021; Outram, 2025).

Geospatial intelligence addresses perimeter deficiencies: The utilisation of autonomous Unmanned Aerial Vehicle (UAV) swarms, powered by Deep Reinforcement Learning (DRL) and complemented by multi-sensor arrays (including satellite imagery, thermal,

and seismic sensors), ensures continuous surveillance over Oman's challenging desert and mountainous boundaries (Lugga, 2025; MDPI Sensors, 2023).

Human supervision mitigates systemic vulnerabilities: Insights from the European Union's Frontex "Smart Borders" project reveal that excessive dependence on automated systems can lead to automation bias, algorithmic prejudice, and legal ambiguities (Babuta & Oswald, 2019; European Commission, 2024; Gugliotta, 2023). Implementing a rigorous "Human-in-the-Loop" (HITL) operational framework guarantees that automated instruments function as analytical consultants, while human officials maintain exclusive jurisdiction over ultimate legal determinations.

6.1 This research offers both theoretical and practical insights into security studies and criminological policy in the Arab Gulf region.

Conceptual Progression

This study broadens Routine Activity Theory (RAT) (Cohen & Felson, 1979) from a criminological viewpoint to encompass modern digital and autonomous defence contexts. In conventional criminological discourse, a crime transpires when a driven perpetrator and an appropriate target intersect in both space and time in the "lack of an effective guardian.

This study illustrates that in distant or high-capacity border settings, the integration of multi-sensor GEOINT, automated predictive analytics, and autonomous drone swarms functions as a "Digitally Competent Guardian" (Monash University, 2025). Through the provision of ongoing situational awareness and anticipatory threat analysis, this digital sentinel eliminates the perceived spatial and temporal obscurity that criminal organisations rely on for conducting unlawful activities.

Pragmatic Policy Framework for Oman

This document offers the Sultanate of Oman and the Royal Oman Police (ROP) a pragmatic operational framework that connects advanced technical innovation with national sovereignty and legal customs.

Operational Integration: Instructions for establishing a centralised Border Intelligence Fusion Center (BIFC) that connects real-time data streams directly to tactical field units.

Capacity Enhancement: Suggestions for advancing the skills of Omani security staff at the Sultan Qaboos Academy for Police Sciences in data literacy, dashboard analysis, and ethical judgement.

Institutional Cohesion: Guaranteeing that AI border instruments function within national cloud frameworks, safeguarding state data sovereignty while achieving the governance objectives of Oman Vision 2040 (NCSI, 2024).

6.2 This document proposes four fundamental policy recommendations for the Omani security apparatus, grounded in the amalgamation of technological data, criminological theory, and comparative case study.

Implement Compulsory "Human-in-the-Loop" Operational Framework: Implement legal provisions within the ROP stipulating that predicted risk scores and automated anomaly notifications function solely as tools for investigation. Conclusive determinations about physical inspections, border refusals, or legal incarcerations must necessitate the approval of a human official.

Ensure National Data Sovereignty: Ascertain that all predictive analytics systems, biometric repositories, and sensor integration networks are only operated on secure domestic cloud infrastructure under Omani jurisdiction, thwarting international third-party data misuse.

Develop Sophisticated Educational Programs: Implement targeted training initiatives at the Sultan Qaboos Academy for Police Sciences concentrating on operational data analysis, algorithmic clarity, and mitigating automation bias among leadership staff.

Establish Ethical Oversight: Formulate the Omani Security AI Governance & Oversight Committee (SAIGOC) to perform semi-annual evaluations of implemented machine learning models, guaranteeing that algorithms are transparent, impartial, and consistent with Omani legal standards.

6.3 As governmental security agencies implement predictive technologies, international organised criminal syndicates will perpetually modify their tactics. Future criminological and technological investigations should concentrate on nascent threat vectors.

Adversarial Machine Learning and Data Contamination: Examining the potential for organised crime groups to intentionally distort input data (such as embedding fraudulent travel patterns or modified cargo documentation into training datasets) to induce algorithmic vulnerabilities or trigger widespread false alerts (Barman & Barman, 2026).

Generative AI and Synthetic Identity Fraud: Evaluating the potential exploitation of generative AI models by criminal organisations to create hyper-realistic synthetic identities, counterfeit documents, or altered biometric credentials intended to deceive automated API/PNR verification systems (UNODC, 2024).

Counter-Unmanned Aerial Systems (C-UAS): Examining strategies to neutralise autonomous smuggling drones or drone swarms utilised by criminal organisations to disrupt border sensors or convey valuable illicit goods over desolate border areas (MDPI Sensors, 2023; Punjab State Open University, 2026).

Through the integration of technical advancements with proactive criminological studies and robust governance, the Sultanate of Oman can develop a contemporary, autonomous,

and ethically principled border defence mechanism that effectively addresses current and impending security challenges.

7. *[DATA INGESTION & SENSOR LAYER]*

Structured Records: PNR, API, Customs Manifests, Immigration Watchlists
Spatial Feeds: Satellite Imagery, Thermal Sensors, Ground Seismic Arrays
Maritime Feeds: Coastal Radar, AIS Tracking, Marine Patrol Logs

[TIER 1: AI ANALYTICS & SIMULATION ENGINE]

supervised Machine Learning: Anomaly Detection & Probabilistic Threat Scoring
Deep Reinforcement Learning (DRL): Autonomous UAV Swarm Route Optimization
Predictive Simulation: 12–48 Hour Spatial Risk Threat Mapping

[TIER 2: RISK EVALUATION & THRESHOLD CHECK]

Low Threat Score = Automated Clearance & Fast-Track Entry
Elevated Threat Score / Anomaly Alert Triggered

[TIER 3: HUMAN-IN-THE-LOOP (HITL) AUDIT CENTER] (Specialized ROP Intelligence Analyst Verification)

Review Feature Attribution & Anomaly Reasoning (Eliminating Black-Box Opacity)
Verify Legal Grounds & Compliance with Omani Statutory Frameworks
Filter Out Automation Bias / Sensor Artifact False Positives
Issue Formally Signed Tactical Interdiction Order

[TIER 4: TACTICAL FIELD INTERDICTION & ACTION]

Land Checkpoints: Directed Physical & Cargo Scanning
Unpopulated Perimeters: Autonomous UAV Swarm Interception & Patrol Dispatch
Maritime Borders: Royal Navy of Oman / Coast Guard Patrol Interdiction

[GOVERNANCE & CONTINUOUS AUDIT LOOP] Security AI Governance & Oversight Committee (SAIGOC)

Bi-annual Algorithmic Health & Bias Audits
Feedback Log: Feeding Valid Interdictions Back into System Training

Audit Compliance with Oman Vision 2040 & Data Sovereignty Mandates

Operational Parameter	Frontex "Smart Borders" (European Union)	Proposed ROP "Smart Border Framework" (Sultanate of Oman)	Criminological & Strategic Significance
Primary System Architecture	Decentralized multi-database interoperability across member states (ETIAS, VIS, EES, CIR).	Centralized Border Intelligence Fusion Center (BIFC) under unified ROP command.	Centralized architecture prevents multi-agency delays and enforces unified operational standards.
Geographic Focus	Mixed land checkpoints, international airports, and Mediterranean maritime coastlines.	High-throughput ports, rugged desert perimeters (Rub' al Khali), mountain zones (Dhofar), and coastal straits (Hormuz).	Tailor-made for Oman's specific terrain where physical border posts are impractical.
Primary Screening Modality	Pre-travel electronic authorization (ETIAS) and automated biometric entry/exit logging.	Integrated PNR/API behavioral anomaly scoring paired with real-time GEOINT sensor fusion.	Shifts focus from static identity checking to predictive behavioral pattern analysis.
Perimeter Surveillance Technology	Satellite monitoring, fixed thermal towers, and piloted aerial patrols.	Autonomous UAV swarms powered by Deep Reinforcement Learning (DRL) and ground seismic sensor arrays.	Eliminates coverage gaps over vast unpopulated terrains, acting as a "Digitally Competent Guardian".
Decision-Making Oversight	Automated pre-screening with manual escalation;	Mandatory Human-in-the-Loop (HITL)	Preserves judicial independence and ensures

	legal vulnerabilities surrounding "black-box" opaque denials.	protocol for every elevated risk alert.	accountability under Omani law (Mas'uliyyah).
Data Sovereignty & Privacy	Complex regional data-sharing across 27 member states under GDPR oversight.	100% sovereign cloud infrastructure hosted strictly within Oman under ROP governance.	Prevents foreign data harvesting, protects national security intelligence, and supports Oman Vision 2040.
Ethical & Bias Mitigation	Ex-post judicial challenges before European courts regarding discriminatory profiling.	Proactive bi-annual algorithmic auditing by the independent SAIGOCcommittee.	Prevents automated systemic bias before deployment, ensuring compliance with UNODC standards.

9. ***AlgorithmWatch. (2025). Algorithmic policing: When predicting means presuming guilty. https://algorithmwatch.org/en/algorithmic-policing-explained/***

Babuta, A., & Oswald, M. (2019). Data analytics and algorithmic bias in policing (RUSI Occasional Paper). Royal United Services Institute for Defence and Security Studies / UK Centre for Data Ethics and Innovation.

https://assets.publishing.service.gov.uk/media/5d7f6b2540f0b61ccdfa4b80/RUSI

Report-_Algorithms_and_Bias_in_Policing.pdf

Barman, P. K., & Barman, P. K. (2026). Unmasking algorithmic bias in predictive policing: A GAN-based simulation framework with multi-city temporal analysis (arXiv Preprint arXiv:2603.18987). arXiv. https://arxiv.org/abs/2603.18987

Barrett, B. (2017). Reasonably suspicious algorithms: Predictive policing at the United States border. NYU Review of Law & Social Change, 41(3), 327–362. https://socialchangenyu.com/wp-content/uploads/2017/09/barrett_digital_9-6-17.pdf

Brouwer, E. (2020). Large-scale database interoperability and the future of EU border governance. European Journal of Migration and Law, 22(4), 479–508. https://doi.org/10.1163/15718166-12340088

Cohen, L. E., & Felson, M. (1979). Social change and crime rate trends: A routine activity approach. American Sociological Review, 44(4), 588–608. https://doi.org/10.2307/2094589

Decision Lab. (2024). Automation bias. https://thedecisionlab.com/biases/automation-bias

European Commission. (2024). Regulation (EU) 2024/1689 of the European Parliament and of the Council laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). Official Journal of the European Union.

European Institute of Management and Technology [EIMT]. (2026). What is automation bias in AI security. https://www.eimt.edu.eu/what-is-automation-bias-in-ai-security

European Union Agency for Fundamental Rights [FRA]. (2018). Under watchful eyes: Biometrics, EU IT systems and fundamental rights. Publications Office of the European Union. https://fra.europa.eu/en/publication/2018/under-watchful-eyes-biometrics-eu-it-systems-and-fundamental-rights

Filho, C. I. (2025). Artificial intelligence in European border policing: Legal challenges, migration governance, and security sector reform. Rocznik Integracji Europejskiej, 19, 135–152. https://rie.amu.edu.pl/wp-

content/uploads/2025/12/RIE-19-2025-08.pdf

Gugliotta, L. (2023). Will AI 'subtly' take over decision-making in the EU migration context? Warnings and lessons from ETIAS and VIS. European Law Review, 48(4), 412–435.

IRIS. (2026). Building a European geospatial intelligence capability: Mapping initiatives, identifying gaps, and outlining strategic options (ARES Policy Paper). Institut de Relations Internationales et Stratégiques. https://www.iris-france.org/wp-content/uploads/2026/01/ARES_2026_01_122_European_Spatial_Intelligence_PolicyPaper.pdf

Kostov, I. (2021). Machine learning and the legal framework for the use of passenger name record data. ResearchGate / European PNR Directive Legal Studies.

https://www.researchgate.net/publication/350054774_Machine_Learning_and_the_Legal_Framework_for_the_Use_of_Passenger_Name_Record_Data

Lugga, S. (2025). Integrating artificial intelligence (AI) with geographic information systems (GIS) and remote sensing technologies for security management. Direct Research Journal of Engineering and Information Technology, 13, 1–10.

https://journals.directresearchpublisher.org/index.php/drjeit/article/download/304/ 272/531

MDPI Sensors. (2023). Application of deep reinforcement learning to UAV swarming for ground surveillance. Sensors, 23(21), Article 8766.

https://doi.org/10.3390/s23218766

Monash University. (2025). Cyber routine activity theory. In Oxford Research Encyclopedia of Criminology and Criminal Justice. Oxford University Press. https://doi.org/10.1093/acrefore/9780190264079.013.782

National Center for Statistics and Information [NCSI]. (2024). Oman digital transformation and governance indicators. Sultanate of Oman.

Open Research Europe. (2026). Machine learning techniques applied in border crossing. Open Research Europe, 6, Article 64. https://doi.org/10.12688/openreseurope.18412.1

Outram, M. (2025). Modern border management: Leveraging AI systems for strategic advantage. World Customs Journal, 19(1), 15–32.

https://www.worldcustomsjournal.org/article/159458-modern-border-management-leveraging-ai-systems-for-strategic-advantage

Punjab State Open University. (2026). Artificial intelligence-enabled autonomous defense drones for real-time threat detection and response. International Journal of AI in Practice, 1(1), 1–12. https://psou.ac.in/beta/AI_in_Practice/articles/view/Paper_AI.pdf

SAS Institute. (2021). Identify high-risk travelers faster and more accurately through better analysis of PNR data(Solution Brief). SAS Campus Drive.

https://www.sas.com/content/dam/SAS/en_us/doc/solutionbrief/identify-high-risk-travelers-pnr-data-109223.pdf

Semantic Scholar Research. (2026). A multi-agent reinforcement learning framework for drone swarm border surveillance. https://pdfs.semanticscholar.org/e979/bb6b0d5cadee26785df3d82a45fd44c1d3fb.pdf

SITA. (2023). Securing the border: SITA intelligence and targeting positioning paper. SITA Border Management White Paper. https://www.sita.aero/globalassets/docs/white-papers/sita-intelligence-and-targeting-positioning-paper.pdf

Travizory Border Security. (2024). Understanding how API-PNR data enhances border security. https://www.travizory.com/understanding-how-api-pnr-data-enhances-border-security

United Nations Office on Drugs and Crime [UNODC]. (2024). Transnational organized crime threat assessment: Evolution of global illicit networks. United Nations Publications. https://www.un.org/en/peace-and-security/transnational-crime

Vavoula, N. (2020). Cyber security and data protection at the EU external borders: The interoperability of EU information systems. Computer Law & Security Review, 37, Article 105413. https://doi.org/10.1016/j.clsr.2020.105413

Završnik, A. (2020). Algorithmic policing in Europe: Challenges and remedies. Eucrim - The European Criminal Law Periodical, 2020(2), 115–122.

https://doi.org/10.26352/E202002115

Vavoula, N. (2020). Cyber security and data protection at the EU external borders: The interoperability of EU information systems. Computer Law & Security Review, 37, Article 105413. https://doi.org/10.1016/j.clsr.2020.105413

Završnik, A. (2020). Algorithmic policing in Europe: Challenges and remedies. Eucrim - The European Criminal Law Periodical, 2020(2), 115–122.

https://doi.org/10.26352/E202002115